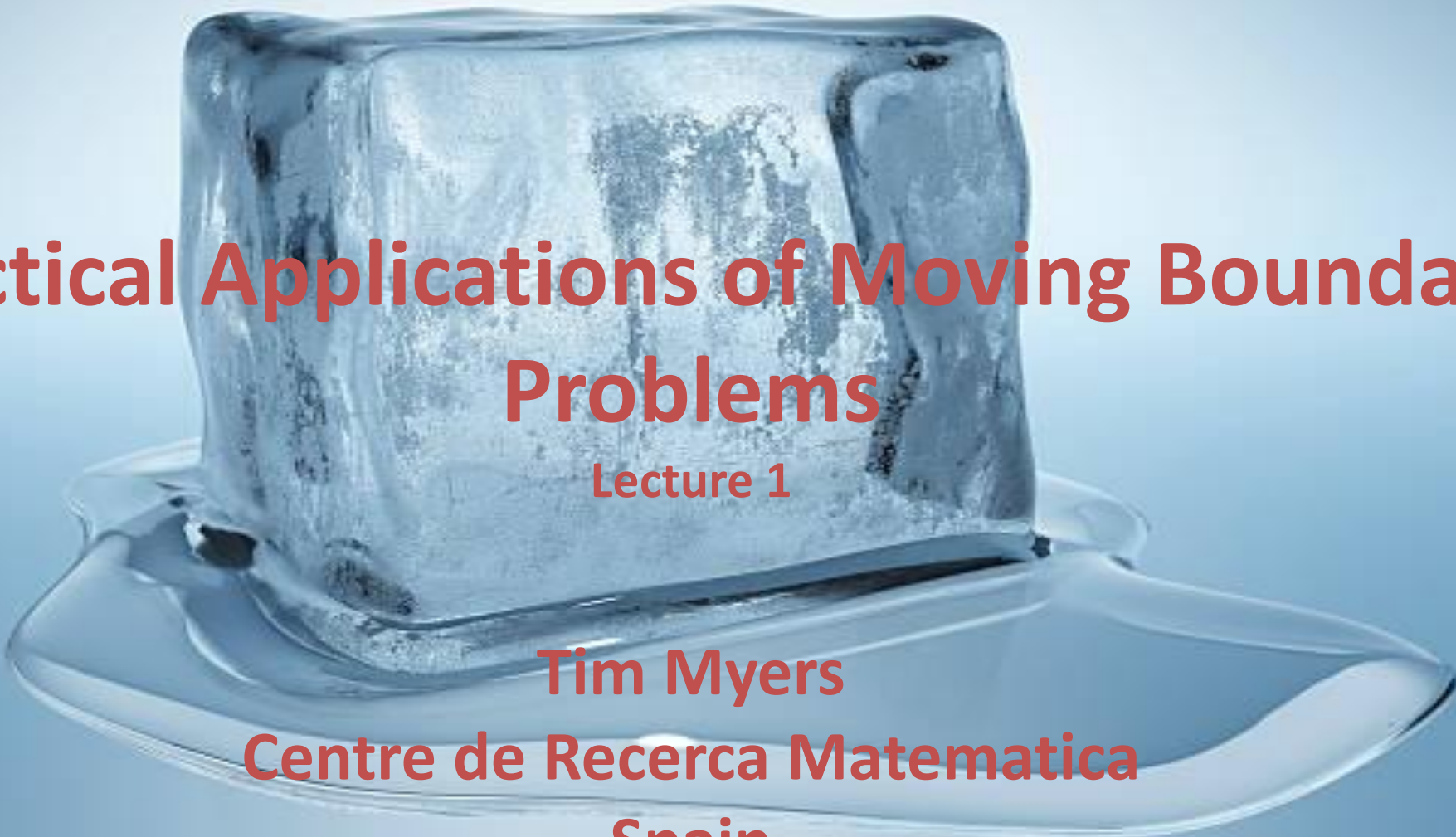




Practical Applications of Moving Boundary Problems

Lecture 1

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What are we studying ...

Problems where the solution of a differential equation has to satisfy certain boundary conditions on a pre-defined domain are known as *Boundary Value Problems* (BVPs). In certain cases this boundary is not known and is determined as part of the solution. If the unknown boundary is fixed, this is termed a Free Boundary Problem. If the boundary is moving we have a *Moving Boundary Problem*. (Certain authors will cover all cases with the term *Free Boundary Problem*).

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Wave boundaries



Growing crystals



Actually ...

We will be more specific and look at problems where the boundary motion is caused by some change of state, for example melting or solidification. These are termed Stefan problems.

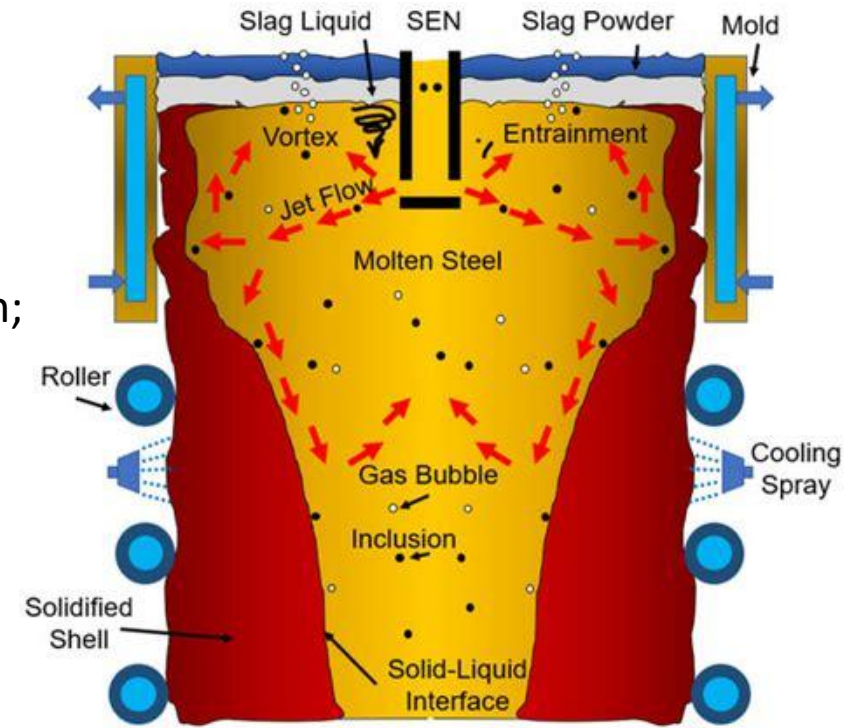
Analogous problems occur in flow in porous media, mathematical finance, shock waves in gas dynamics, contaminant removal from liquids, crack growth in solids and polymer and grain swelling. Aircraft icing.

To keep the notation simple and consistent we will work in terms of phase change (melting, evaporation etc)



Melting ice;
Melting; Solidification;
Evaporation; Condensation;
Ablation

+ mathematical analogies



Aircraft icing



Structure of the course

Lecture 1: Can't solve a phase change problem without studying ... The Heat Equation and ways to analyse it.

Lecture 2: We're now ready for the Stefan problem ... exact and approximate solutions.

Lecture 3: Problems with flow – thin fluid layers, aircraft icing.

Lecture 4: Solidification with varying properties – supercooled, nanoscale.

Lecture 5: Let's see how 1-4 go.

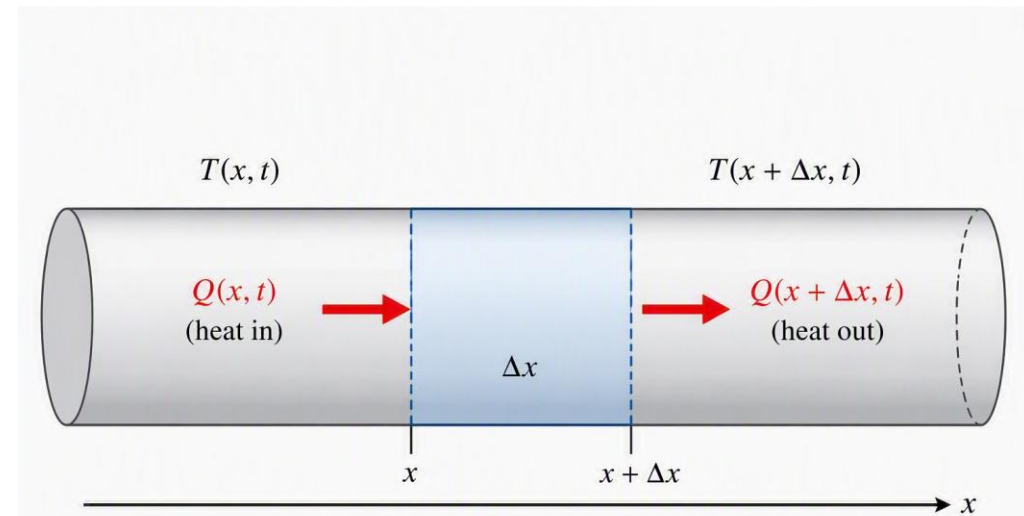
Derivation of the heat equation

The one-dimensional heat equation is derived by combining the **conservation of thermal energy** with **Fourier's law of heat conduction**.

- Consider a thin metal rod of cross-sectional area A .
- Look at a small segment of the rod between position x and $x + \Delta x$.
- The total heat energy E inside this small segment depends on its temperature $T(x, t)$, density ρ , and specific heat capacity c :

$$\text{Energy} = c\rho T(x, t) \cdot (A\Delta x)$$

Difference between heat and temperature?



Rate of change

- The rate of change of this heat energy over time is:

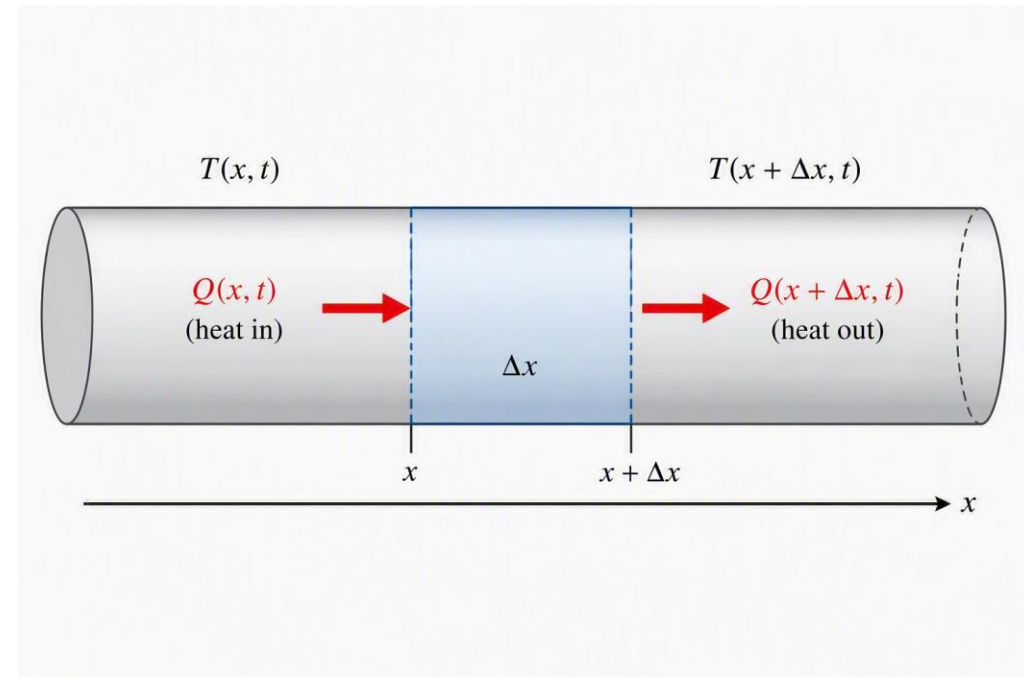
$$\frac{\partial}{\partial t}[c\rho T(x, t)]A\Delta x$$

- This change equals the net heat flowing into the segment from the boundaries (heat flux Q entering at x minus heat flux leaving at $x + \Delta x$) multiplied by the area A :

$$c\rho \frac{\partial T}{\partial t} A\Delta x = Q(x)A - Q(x + \Delta x)A$$

- Divide both sides by $A\Delta x$ and take the limit as $\Delta x \rightarrow 0$ to get:

$$c\rho \frac{\partial T}{\partial t} = -\frac{\partial Q}{\partial x}$$

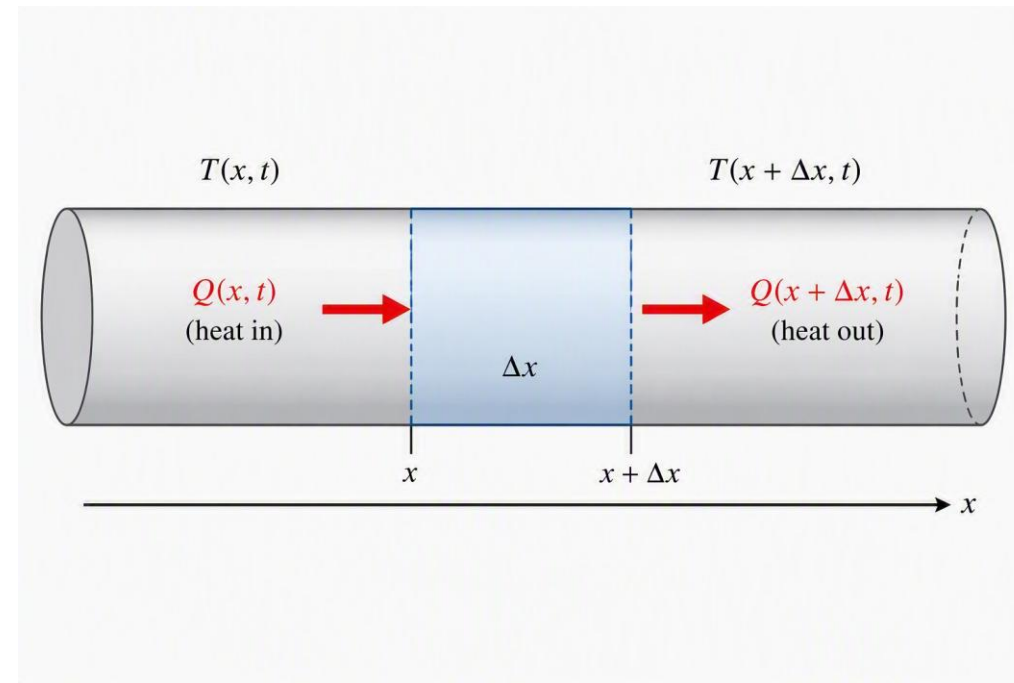


Fourier's law

- **Fourier's law** states that heat flows from hot to cold, and the rate of heat flow Q is proportional to the temperature gradient:

$$Q = -k \frac{\partial T}{\partial x}$$

where k is the thermal conductivity.



The Heat Equation!

- Substitute Fourier's law into the energy balance equation:

$$c\rho\frac{\partial T}{\partial t} = -\frac{\partial}{\partial x} \left(-k\frac{\partial T}{\partial x} \right)$$

- Assuming c , ρ , and k are constant throughout the rod, we pull k out of the derivative:

$$c\rho\frac{\partial T}{\partial t} = k\frac{\partial^2 T}{\partial x^2}$$

- Divide both sides by $c\rho$ to define the **thermal diffusivity** $\alpha = \frac{k}{c\rho}$:

$$\frac{\partial T}{\partial t} = \alpha\frac{\partial^2 T}{\partial x^2}$$

The Heat Equation!

The heat equation is an example of a Conservation Law - in this case conservation of energy

$$\frac{\partial E}{\partial t} + \frac{\partial}{\partial x} Q = 0$$

where the energy $E = \rho c T$ and flux $Q = -k T_x$

In continuum mechanics, the most general form of an exact conservation law is given by a continuity equation

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x} Q = 0$$

where ρ is the density and Q the flux

In fluid mechanics ρ is the fluid density $Q = \rho u$

Electricity ρ is the charge density

Kinematic wave $Q = -Ch + Dh_x$, where h is wave height

Traffic flow ...

The Heat Equation!

1st order in time, 2nd order in space => 1 initial condition, 2 boundary conditions

Initial condition just = the initial state, i.e. the temperature profile at the start

Boundary conditions ...

- Specified temperature (Dirichlet condition)

$$T = T_0 .$$

- Specified heat flux (Neumann condition)

$$k \frac{\partial T}{\partial x} = -q .$$

- Heat flux proportional to a temperature difference (Newton cooling or Robin condition)

$$k \frac{\partial T}{\partial x} = \mathcal{H}(T_a - T)$$

Radiation –

Flux proportional to T^4

Other forms

- More dimensions

$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T$$

- With flow

$$\left(\frac{\partial T}{\partial t} + \mathbf{u} \cdot \nabla T \right) = \alpha \nabla^2 T$$

- With a source (friction/shear heating, chemical reaction ...)

$$\frac{\partial T}{\partial t} = \alpha \nabla^2 T + f(x, t, T)$$

Variable thermal coefficients ...

Other forms

Perfume, tea, oxygen into blood, CO2 leaving blood, fluids mixing

- Diffusion equation

$$\frac{\partial C}{\partial t} = D \nabla^2 C$$

- Random walk/Brownian motion

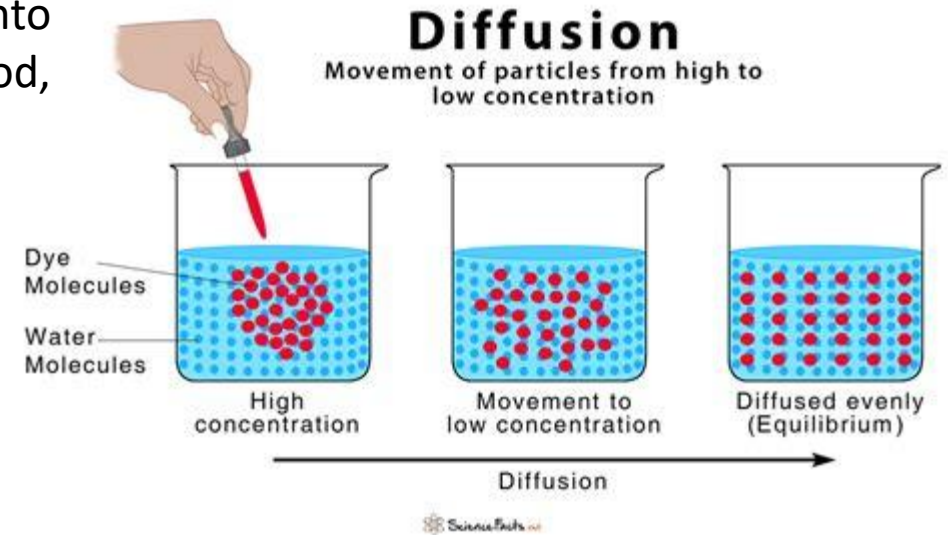
$$\frac{\partial p}{\partial t} = D \nabla^2 p$$

where p is the probability density $p(x,t)$ of a Brownian particle.

- Black-Scholes

$$\frac{\partial V}{\partial t} = -\frac{1}{2}\sigma^2 S^2 \nabla^2 V - rS \frac{\partial V}{\partial S} + rV$$

V is the price of the option as a function of stock price S and time t , r is the risk-free interest rate and σ is the volatility of the stock (Backwards heat equation with a source)



Other forms

In the complex domain (and backwards)

The Schrodinger equation is a partial differential equation that governs the wave function of a non-relativistic quantum-mechanical system

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi$$

where Ψ is the wave function, m is the mass of the particle, and V is the potential energy function that represents the environment in which the particle exists.

Basically the heat/diffusion equation is everywhere

Solving the heat equation - Separation of variables/Fourier series

This is the most common method of solving the 1D heat equation ...

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

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$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

No use to us!!!

Similarity solutions

We seek a self-similar solution of the form

$$T(x, t) = f(\eta) \quad \text{where} \quad \eta = \frac{x}{2\sqrt{\alpha t}}$$

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

Using the chain rule,

$$T_t = f'(\eta)\eta_t, \quad T_x = f'(\eta)\eta_x.$$

Then

$$\eta_t = -\frac{x}{4\sqrt{\alpha} t^{3/2}} = -\frac{\eta}{2t}, \quad \Rightarrow T_t = -\frac{\eta}{2t} f'(\eta).$$

$$\eta_x = \frac{1}{2\sqrt{\alpha t}}, \quad \Rightarrow T_x = \frac{f'(\eta)}{2\sqrt{\alpha t}}, \quad T_{xx} = \frac{f''(\eta)}{4\alpha t}$$

Similarity solutions

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

Heat equation in similarity form

$$-\frac{\eta}{2t} f' = \alpha \left(\frac{f''}{4\alpha t} \right) \Rightarrow f'' + 2\eta f' = 0$$

No time

Let

$$g = f' \Rightarrow g' + 2\eta g = 0$$

This is separable

$$\frac{dg}{g} = -2\eta d\eta \Rightarrow \ln g = -\eta^2 + C_1 \Rightarrow g = Ce^{-\eta^2}$$

Since $g = f'$,

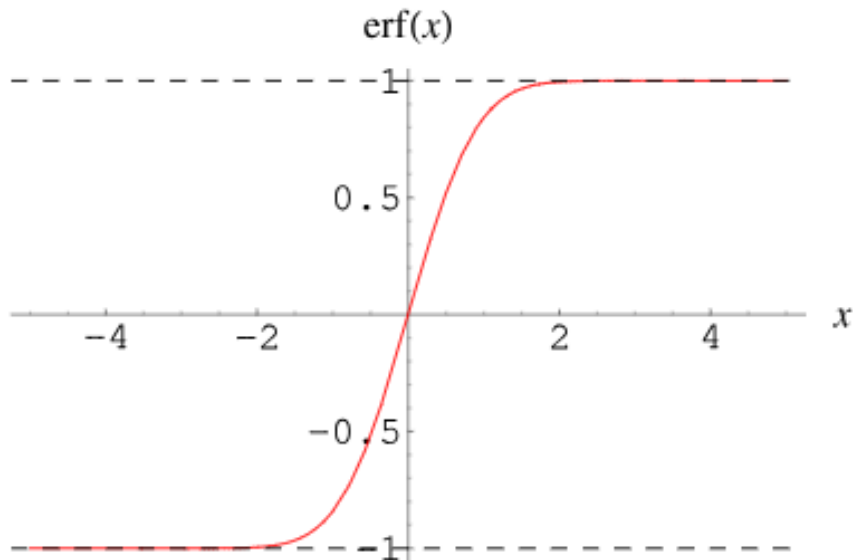
$$f(\eta) = C \int_0^\eta e^{-s^2} ds + D.$$

The *Error Function* is defined as

$$\operatorname{erf}(\eta) = \frac{2}{\sqrt{\pi}} \int_0^\eta e^{-s^2} ds,$$

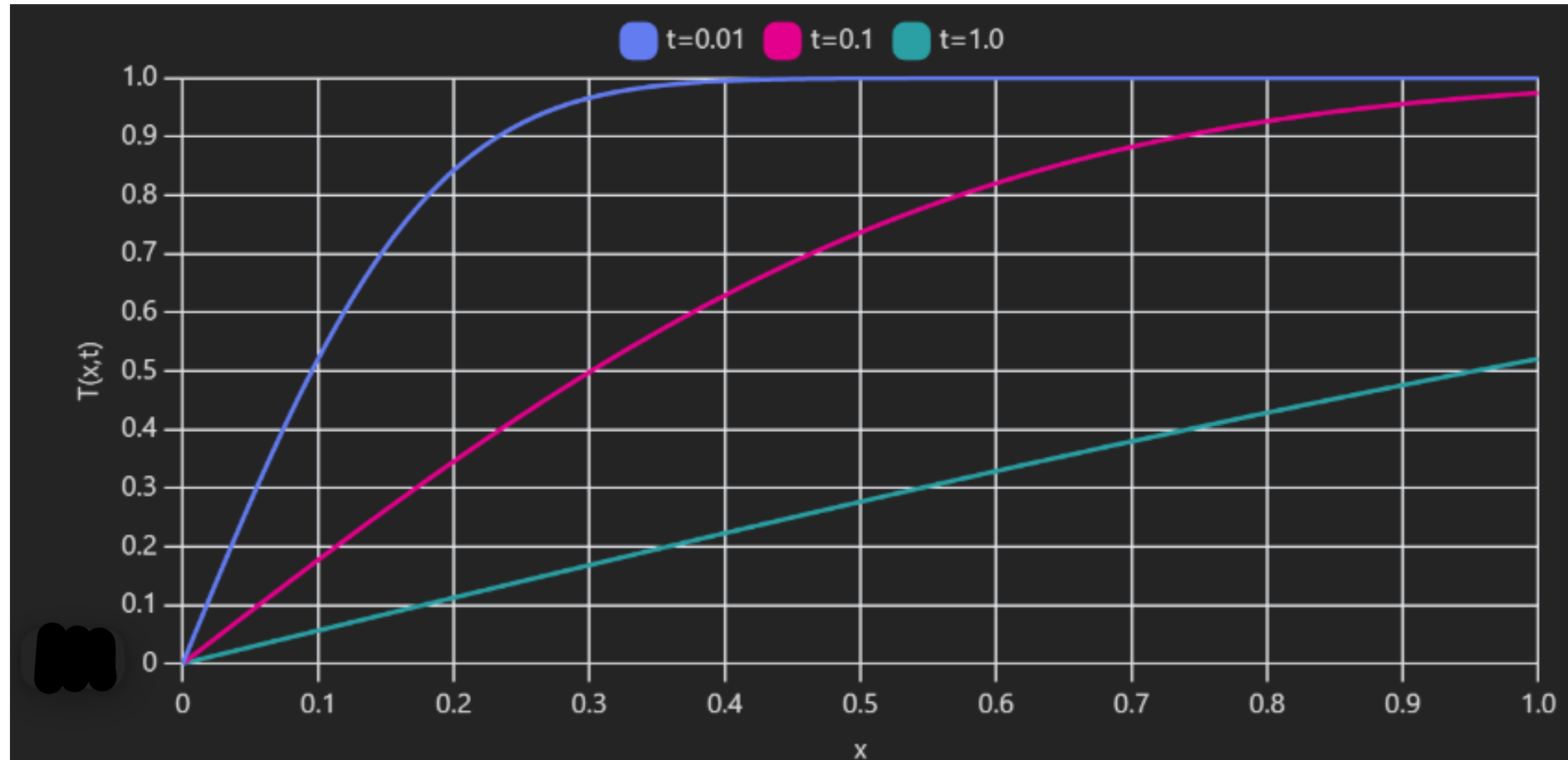
Hence

$$f(\eta) = A \operatorname{erf}(\eta) + B \Rightarrow T(x, t) = A \operatorname{erf}\left(\frac{x}{2\sqrt{\alpha t}}\right) + B$$



Similarity solutions

Error function solution of the heat equation with $\alpha=1$ and $T(0,t)=0$, $T(x,0)=1$, $T=1$ at infinity



Due to the form of the similarity variable it is hard to specify x co-ordinates (eta becomes function of t), except 0 and infinity So why did I show it???

Similarity solutions

Error function solution of the heat equation with $\alpha=1$ and $T(0,t)=0$, $T(x,0)=1$, $T=1$ at infinity ---

$$T(x, t) = A \operatorname{erf} \left(\frac{x}{2\sqrt{\alpha t}} \right) + B$$

$$A=1, B=0$$

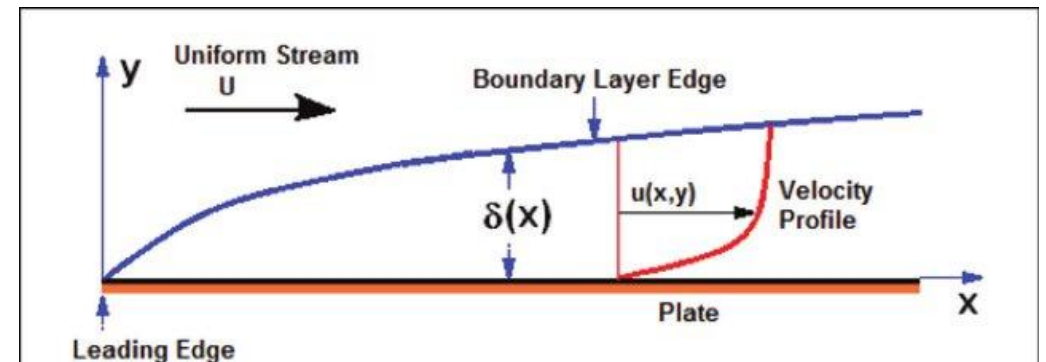
Two BCs and 1 IC – determine A, B through the BCs

The solution works because

$$\eta = \frac{x}{2\sqrt{\alpha t}}$$

and the condition at infinity x matches the initial condition (so only need 2 conditions)
– it is not so unusual but is still a strong restriction

Check out the classic Blasius boundary layer



Fundamental solutions

Can also derive a family of solutions from the error function

Derivatives also satisfy the heat equation

$$T_t - \alpha T_{xx} = 0 \quad \frac{\partial}{\partial x} (T_t - \alpha T_{xx}) = 0$$

$$\frac{\partial T_x}{\partial t} = \alpha \frac{\partial^2 T_x}{\partial x^2} \Rightarrow T_x \text{ is a solution}$$

$$\frac{\partial T_t}{\partial t} = \alpha \frac{\partial^2 T_t}{\partial x^2} \Rightarrow T_t \text{ is a solution}$$

and so on with other derivatives - hence *Fundamental solutions*

Note form of solution variable

$$T_x = \theta(x, t) = \frac{e^{-x^2/(4\alpha t)}}{\sqrt{\pi\alpha t}}$$

$$T_t = \theta(x, t) = -\frac{\alpha x e^{-x^2/(4\alpha t)}}{2\sqrt{\pi}(\alpha t)^{3/2}} \quad \text{etc}$$

Non-dimensionalisation

In the real world parameter values, BCs, ICs are varied

In mathematics classes parameter values etc are usually nice numbers
(I chose 0, 1 in previous example)

Why??? Because problems are usually presented in
non-dimensional form

$\alpha \ll 1$?

Substance	T_m (K)	L_m (J/Kg)	c_l, c_s (J/Kg·K)	ρ_l, ρ_s (kg/m ³)	k_l, k_s (W/m·K)	σ_{sl} (N/m)
Water	273.15	3.34×10^5	4181, 2050	1000, 920	0.55, 2.20	0.03
Gold	1337	6.37×10^4	163, 129	17,300, 19,300	106, 317	0.27
Lead	600	2.30×10^4	148, 128	10,700, 11,300	16,35	0.05
Tin	505	5.85×10^4	268, 230	6980, 7180	30, 67	0.064
Copper	136	2.09×10^5	530, 385	8020, 8960	157.2, 413	0.182
Salol	316	9×10^4	1580, -	1180, 1250	0.18, -	-
n-Octadecane	301.3	2.435×10^5	2161, 2151	771, 930	0.15, 0.372	-

Non-dimensionalisation

We find 1000 birds on a remote island, the maximum number of birds the resources can sustain (the carrying capacity) is 6000. After one month the population has increased to 1125 birds.
Population modelled by a logistic equation

$$\frac{dx}{dt} = r \left(1 - \frac{x}{K} \right) x$$

where x is the number of birds, r the growth rate and K the carrying capacity
Introduce dimensionless variables $\hat{x} = x/X$, $\hat{t} = t/\tau$ X and τ are (constant) typical values

$$\frac{d(X\hat{x})}{d(\tau\hat{t})} = r \left(1 - \frac{X\hat{x}}{K} \right) X\hat{x}$$

Choose $X = K$

$$\frac{d\hat{x}}{d\hat{t}} = r\tau (1 - \hat{x}) \hat{x}$$

Choose $\tau = 1/r$

Expect $\hat{x}$ and its derivatives to have size around unity

Initial condition $x(0) = 1000 \Rightarrow \hat{x}(0) = \hat{x}_0 = 1000/6000 = 1/6$

Problem reduced from one depending on three parameters to depending on one parameter

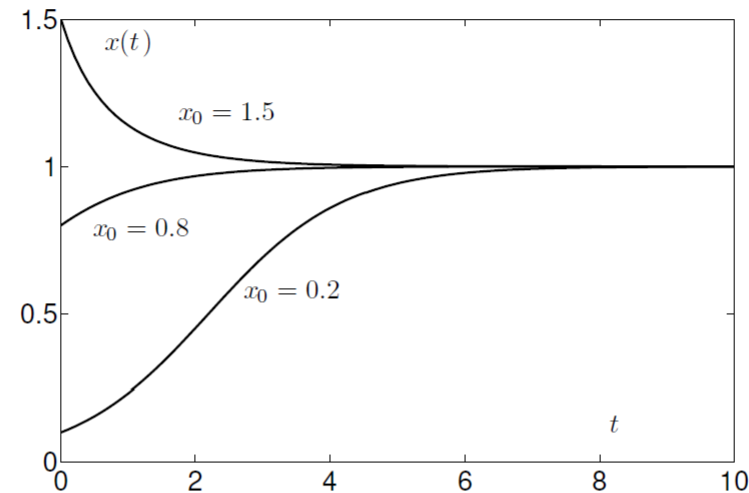
Reduced parameter
space

Non-dimensionalisation

Solution is

$$\hat{x} = \frac{\hat{x}_0}{\hat{x}_0 + (1 - \hat{x}_0)e^{-\hat{t}}}$$

One-parameter family of solutions



Non-dimensional bird population for $\hat{x}_0 = 0.2, 0.8, 1.5$

Dimensional result – replace $\hat{t}$ with $t/T = rt$ and $\hat{x}$ with $x/X = x/K$ (value of $r = 0.14$ found by setting $x = 1125$ at $t = 1$, $\hat{x} = 1125/6000$, $\hat{t} = r$)

$$\frac{x}{K} = \frac{x_0}{x_0 + (K - x_0)e^{-rt}}$$

Non-dimensionalisation

This is not an exact science ... Generally I want to scale such that all terms are order unity and coefficients are order unity or smaller, we may use the physics to inform the scaling or perhaps a region of interest.

Another possible scaling based on initial condition

Could scale x with x_0

$$\hat{x} = \frac{x}{K} \quad \text{or} \quad \hat{x} = \frac{x}{x_0}$$

Scaling not always unique (and not always obvious)

First scaling if interest lies in long time behaviour

Second for small times, when x is close to x_0

Setting $X = x_0$

$$\frac{d\hat{x}}{d\hat{t}} = \left(1 - \frac{\hat{x}}{\alpha}\right) \hat{x}$$

where $\alpha = K/x_0$

The governing equation has a parameter α , but the initial condition is simply $\hat{x}(0) = 1$

Non-dimensionalisation

Now to the heat equation ...

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

such that

$$T(x, 0) = T_i \quad T(0, t) = T_0 \quad T(L, t) = T_L$$

Change variables such that

$$\hat{T} = \frac{T - T_0}{T_L - T_0} \quad \hat{x} = \frac{x}{L} \quad \hat{t} = \frac{t}{\tau} \quad \hat{T} = \frac{T - T_0}{T_L - T_0}$$

$$\frac{\partial(\Delta T \hat{T})}{\partial(\tau \hat{t})} = \alpha \frac{\partial^2(\Delta T \hat{T})}{\partial(\mathcal{L} \hat{x})^2} \Rightarrow \frac{\partial \hat{T}}{\partial \hat{t}} = \frac{\alpha \tau}{\mathcal{L}^2} \frac{\partial^2 \hat{T}}{\partial \hat{x}^2}$$

Non-dimensionalisation

$$\frac{\partial \hat{T}}{\partial \hat{t}} = \frac{\alpha \tau}{\mathcal{L}^2} \frac{\partial^2 \hat{T}}{\partial \hat{x}^2}$$

Can't scale out
all temps!

Choose $\tau = \mathcal{L}^2 / \alpha$, $\mathcal{L} = L$, $\Delta T = T_0 - T_L$, $T_0 - T_i$?

Interested in what happens

Near $x = 0$?

Near $t = 0$?

At large times?

What if semi-infinite/infinite media?

Implication of time-scale?

Non-dimensionalisation

$$\frac{\partial \hat{T}}{\partial \hat{t}} = \frac{\alpha \tau}{\mathcal{L}^2} \frac{\partial^2 \hat{T}}{\partial \hat{x}^2}$$

Estimate the time-scale for heat flow through solid and liquid gold and solid and liquid water (assuming there is no flow) for a sample of 10cm.

Substance	T_m (K)	L_m (J/Kg)	c_l, c_s (J/Kg·K)	ρ_l, ρ_s (kg/m ³)	k_l, k_s (W/m·K)	σ_{sl} (N/m)
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Answers in the wrong order

266, 76018, 8572, 78.5s

Which is which?